AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Course Title

INTRODUCTION TO DATABASE

FALL 2023-2024

Section: G

TITLE

AIUB CRICKET TEAM MANAGEMENT SYSTEM

Supervised By

MD Sajid Bin Faisal

Submitted By: Group No-13

Name	ID
Bijoy Ahamed	**22-449373-3**

TABLE OF CONTENTS

TOPICS		Page no.
Title Page		1
Table of Content		2
1.	Introduction	3
2.	Case Study	3
3.	ER Diagram	4
4.	Normalization	5-7
5.	Finalization And Final Table	7-8
6.	Table Creation	9-16
7.	Data Insertion	17-20
8.	Query Test	21-25
9.	DB Connection	26
10.	Conclusion	27

INTRODUCTION

A database management system (DBMS) refers to the technology for creating and managing databases. The main objective of this database project is to create a database to centrally handle **AIUB Cricket Team Management**. The database can provide various information about the teams or matters within the teams such as players or coaches of the team. Other matters such as selectors, officials, events and tournaments can be found also. In completing the project oracle 10g and draw.io.

CASE STUDY

In AIUB Cricket Team Management System, every player has a unique ID named P_ID and age named P_Age. Each player has a name which is indicated by P_Name player position named P_Position, Date of birth named P_DOB and playing role as Role a multivalued attribute as skill and address. A player address is composed of city and country. Player indicates a Selection which means he must be confirmed by selectors and has a full official relationship with officials. A player is confirmed by at least one selector. System can track the player who is appointed by officials and confirmed by selectors. System records each player who are selected by officials. Every selected player is corresponded in selectors. Each Selector has a unique id named S_ID and selector named S_Name. Officials have also their official person name as OP_Name, officers unique ID as O_ID Designation as job and Hire date as joining and the place where they assigned as Details which has two composite Area, Zone. A player may be selected by more than one official person or a official person may be select many player. Officials moderate all sort of events and official relationship with player. Officials appoint Coaches. A Coache may be appointed by more than one official person. A Official person may also appoint many Coaches. Each Coach has a unique ID named C_ID and a multivalued attribute as E-mail named C_E_mail. Coaches also have names named C_Name and coaching role as C_Role. Coaches guide players. A player may be guide by at least one Coache. A Coache may also guide many player. When any event is happening, every event has a unique ID named E_ID. Event has a multivalued attribute as location. Every event has a name as E_Name. Events are moderated by officials. An event can also be moderated by at least one official person or a official person can moderate many event. There are different types of tournaments named Test, T20, ODI and where the tournament happens it is indicated as place which contains Stadium and Gate. Every tournament has an unique number as T_No. Tournaments are operated by events. One Tournament may be operated by exactly one event.

ER DIAGRAM

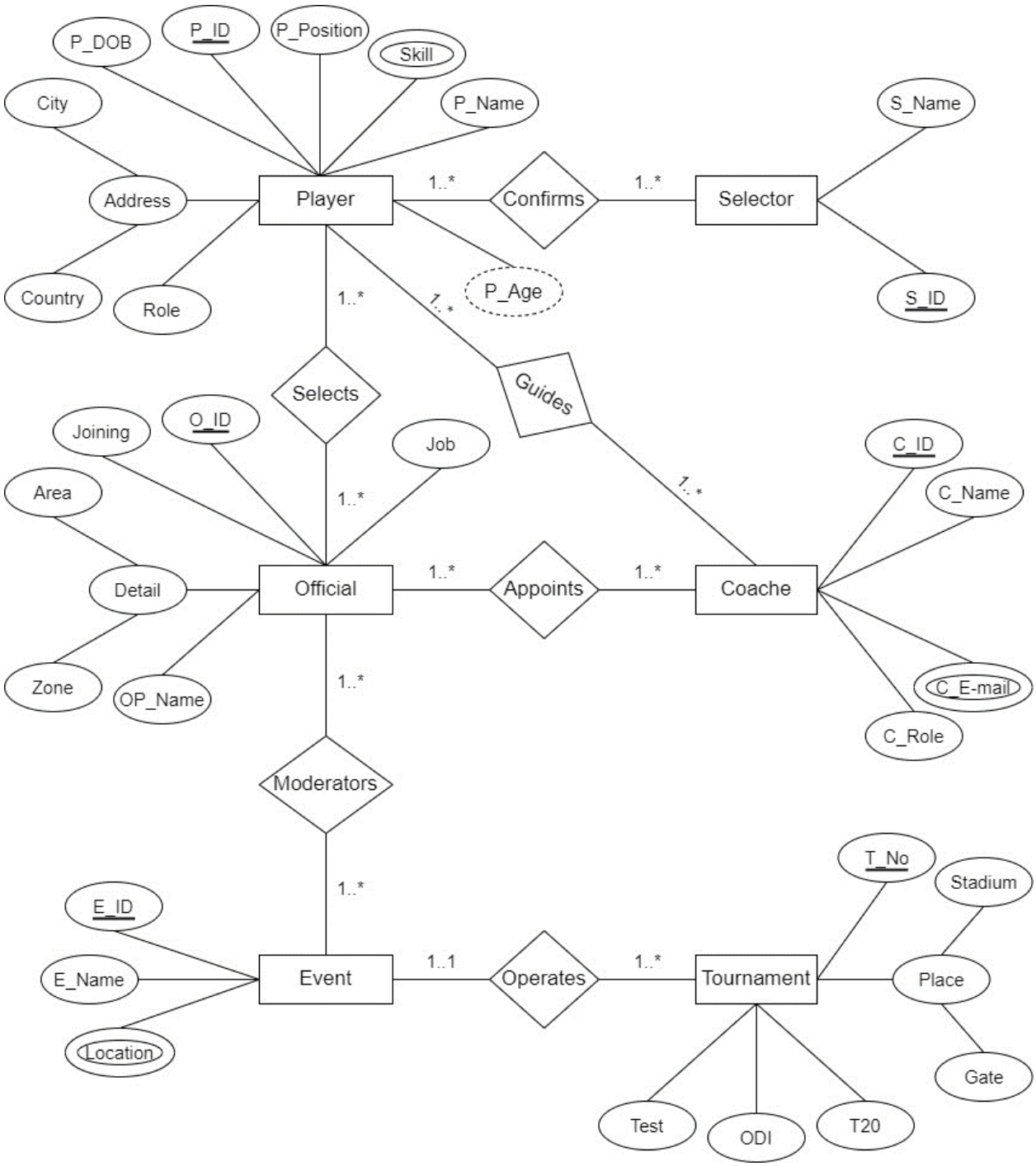

Fig-3.1: ER Diagram.

NORMALIZATION

- **Selector Confirms Player:**

UNF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, S_ID, S_Name.

1NF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, S_ID, S_Name.

2NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City, Country.

2. S_ID(PK), S_Name.

3. P_ID(PK), S_ID(FK).

3NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City.

2. S_ID(PK), S_Name.

3. P_ID(PK), S_ID(FK).

4. City, Country.

- **Official Selects Player:**

UNF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, O_ID, OP_Name, Joining, Job, Detail, Zone, Area.

1NF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, O_ID, OP_Name, Joining, Job, Detail, Zone, Area.

2NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City, Country.

2. O_ID(PK), OP_Name, Joining, Job, Zone, Area.

3. P_ID(PK), O_ID(FK).

3NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City.

2. O_ID(PK), OP_Name, Joining, Job, Zone.

3. P_ID(PK), O_ID(FK).

4. Zone, Area, City, Country.

- **Coache Guides Player:**

UNF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, C_ID, C_Name, C_Role, C_E_mail.

1NF: P_ID, P_Name, P_Position, P_DOB, P_Age, Skill, Role, Address, City, Country, C_ID, C_Name, C_Role, C_E_mail.

2NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City, Country.

2. C_ID(PK), C_Name, C_Role, C_E_mail.

3. P_ID(PK), C_ID(FK).

3NF: 1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City.

2. C_ID(PK), C_Name, C_Role, C_E_mail.

3. P_ID(PK), C_ID(FK).

4. City, Country.

- **Official Appoints Coache:**

UNF: O_ID, OP_Name, Joining, Job, Detail, Zone, Area, C_ID, C_Name, C_Role, C_E_mail.

1NF: O_ID, OP_Name, Joining, Job, Detail, Zone, Area, C_ID, C_Name, C_Role, C_E_mail.

2NF: 1. O_ID(PK), OP_Name, Joining, Job, Zone, Area.

2. C_ID(PK), C_Name, C_Role, C_E_mail.

3. O_ID(PK), C_ID(FK).

3NF: 1. O_ID(PK), OP_Name, Joining, Job, Zone.

2. C_ID(PK), C_Name, C_Role, C_E_mail.

3. O_ID(PK), C_ID(FK).

4. Zone, Area.

- **Official Moderators Event:**

UNF: O_ID, OP_Name, Joining, Job, Detail, Zone, Area, E_ID, E_Name, Location.

1NF: O_ID, OP_Name, Joining, Job, Detail, Zone, Area, E_ID, E_Name, Location.

2NF: 1. O_ID(PK), OP_Name, Joining, Job, Zone, Area.

2. E_ID(PK), E_Name, Location.

3. O_ID(PK), E_ID(FK).

3NF: 1. O_ID(PK), OP_Name, Joining, Job, Zone.

2. E_ID(PK), E_Name, Location.

3. O_ID(PK), E_ID(FK).

4. Zone, Area.

- **Event Operates Tournament:**

UNF: E_ID, E_Name, Location, T_No, Test, ODI, T20, Place, Stadium, Gate.

1NF: E_ID, E_Name, Location, T_No, Test, ODI, T20, Place, Stadium, Gate.

2NF: 1. T_No(PK), Test, ODI, T20, Stadium, Gate, E_ID(FK).

2. E_ID(PK), E_Name, Location.

3NF: 1. T_No(PK), Test, ODI, T20, Stadium, Gate, E_ID(FK).

2. E_ID(PK), E_Name, Location.

FINALIZATION

1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City.
2. S_ID(PK), S_Name.
3. P_ID(PK), S_ID(FK).
4. City, Country.
5. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City. ✕
6. O_ID(PK), OP_Name, Joining, Job, Zone.
7. P_ID(PK), O_ID(FK).
8. Zone, Area, City, Country.
9. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City. ✕
10. C_ID(PK), C_Name, C_Role, C_E_mail.
11. P_ID(PK), C_ID(FK).
12. City, Country. ✕
13. O_ID(PK), OP_Name, Joining, Job, Zone. ✕
14. C_ID(PK), C_Name, C_Role, C_E_mail. ✕
15. O_ID(PK), C_ID(FK).
16. Zone, Area.

17. O_ID(PK), OP_Name, Joining, Job, Zone. ✕

18. E_ID(PK), E_Name, Location.

19. O_ID(PK), E_ID(FK).

20. Zone, Area. ✕

21. E_ID(PK), E_Name, Location. ✕

22. T_No(PK), Test, ODI, T20, Stadium, Gate, E_ID(FK).

FINAL TABLE

1. P_ID(PK), P_Name, P_Position, P_DOB, P_Age, Skill, Role, City. (Player)
2. S_ID(PK), S_Name. (Selector)
3. P_ID(PK), S_ID(FK). (Confirm)
4. City, Country. (Address)
5. O_ID(PK), OP_Name, Joining, Job, Zone. (Official)
6. P_ID(PK), O_ID(FK). (Selects)
7. Zone, Area, City, Country. (Detail_Address)
8. C_ID(PK), C_Name, C_Role, C_E_mail. (Coache)
9. P_ID(PK), C_ID(FK). (Guid)
10. O_ID(PK), C_ID(FK). (Appoint)
11. Zone, Area. (Detail)
12. E_ID(PK), E_Name, Location. (Event)
13. O_ID(PK), E_ID(FK). (Moderator)
14. T_No (PK), Test, ODI, T20, Stadium, Gate, E_ID(FK). (Operates)

TABLE CREATION

1. **Create table Player:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Player (P_ID number (6) primary key, P_Name varchar (20), P_Position varchar (20), P_DOB varchar (20),
                     P_Age varchar (20), Skill varchar (20), Role varchar (20), City varchar (20))
Describe Player
```

Object Type **TABLE** Object **PLAYER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PLAYER	P_ID	Number	-	6	0	1	-	-	-
	P_NAME	Varchar2	20	-	-	-	✓	-	-
	P_POSITION	Varchar2	20	-	-	-	✓	-	-
	P_DOB	Varchar2	20	-	-	-	✓	-	-
	P_AGE	Varchar2	20	-	-	-	✓	-	-
	SKILL	Varchar2	20	-	-	-	✓	-	-
	ROLE	Varchar2	20	-	-	-	✓	-	-
	CITY	Varchar2	20	-	-	-	✓	-	-

1 - 8

Fig-6.1: Player table and Description.

2. **Create table Selector:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Selector (S_ID number (4) primary key, S_Name varchar (20))
Describe Selector
```

Object Type **TABLE** Object **SELECTOR**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SELECTOR	S_ID	Number	-	4	0	1	-	-	-
	S_NAME	Varchar2	20	-	-	-	✓	-	-
									1-2

Fig-6.2: Selector table and Description.

3. **Create table Confirm:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Confirm (P_ID number (6) primary key, S_ID number (4),
       constraint sfk foreign key (S_ID) references  Selector (S_ID))
Describe Confirm
```

Object Type **TABLE** Object **CONFIRM**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CONFIRM	P_ID	Number	-	6	0	1	-	-	-
	S_ID	Number	-	4	0	-	✓	-	-
									1-2

Fig-6.3: Confirm table and Description.

4. **Create table Address:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Address ( Country varchar (20), City varchar (20))
Describe Address
```

Object Type **TABLE** Object **ADDRESS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ADDRESS	COUNTRY	Varchar2	20	-	-	-	✓	-	-
	CITY	Varchar2	20	-	-	-	✓	-	-
									1 - 2

Fig-6.4: Address table and Description.

5. **Create table Official:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Official (O_ID number (5) primary key, OP_Name varchar (25),
                Joining varchar (25), Job varchar (25), Zone varchar (25))
Describe Official
```

Object Type **TABLE** Object **OFFICIAL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
OFFICIAL	O_ID	Number	-	5	0	1	-	-	-
	OP_NAME	Varchar2	25	-	-	-	✓	-	-
	JOINING	Varchar2	25	-	-	-	✓	-	-
	JOB	Varchar2	25	-	-	-	✓	-	-
	ZONE	Varchar2	25	-	-	-	✓	-	-
									1 - 5

Fig-6.5: Official table and Description.

6. **Create table Selects:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Selects (P_ID number (6) primary key, O_ID number (5),
       constraint ofk foreign key (O_ID) references  Official (O_ID))
Describe Selects
```

Object Type **TABLE** Object **SELECTS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SELECTS	P_ID	Number	-	6	0	1	-	-	-
	O_ID	Number	-	5	0	-	✓	-	-
									1 - 2

Fig-6.6: Selects table and Description.

7. **Create table Detail_Address:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Detail_Address (  Area varchar (25), Zone varchar (25),
                               Country varchar (20), City varchar (20))
Describe Detail_Address
```

Object Type **TABLE** Object **DETAIL_ADDRESS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DETAIL_ADDRESS	AREA	Varchar2	25	-	-	-	✓	-	-
	ZONE	Varchar2	25	-	-	-	✓	-	-
	COUNTRY	Varchar2	20	-	-	-	✓	-	-
	CITY	Varchar2	20	-	-	-	✓	-	-
									1 - 4

Fig-6.7: Detail_Address table and Description.

8. **Create table Coache:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Coache (C_ID number (4) primary key, C_Name varchar (20),
                     C_Role varchar (20), C_E_mail varchar (20))
Describe Coache
```

Object Type **TABLE** Object **COACHE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
COACHE	C_ID	Number	-	4	0	1	-	-	-
	C_NAME	Varchar2	20	-	-	-	✓	-	-
	C_ROLE	Varchar2	20	-	-	-	✓	-	-
	C_E_MAIL	Varchar2	20	-	-	-	✓	-	-
									1 - 4

Fig-6.8: Coache table and Description.

9. **Create table Guid:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Guid (P_ID number (6) primary key, C_ID number (4),
       constraint cfk foreign key (C_ID) references  Coache (C_ID))
Describe Guid
```

Object Type **TABLE** Object **GUID**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
GUID	P_ID	Number	-	6	0	1	-	-	-
	C_ID	Number	-	4	0	-	✓	-	-
									1 - 2

Fig-6.9: Guid table and Description.

10. **Create table Appoint:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Appoint (O_ID number (5) primary key, C_ID number (4),
        constraint ccfk foreign key (C_ID) references  Coache (C_ID))
Describe Appoint
```

Object Type **TABLE** Object **APPOINT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
APPOINT	O_ID	Number	-	5	0	1	-	-	-
	C_ID	Number	-	4	0	-	✓	-	-
									1 - 2

Fig-6.10: Appoint table and Description.

11. **Create table Detail:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Detail ( Area varchar (25), Zone varchar (25))
Describe Detail
```

Object Type **TABLE** Object **DETAIL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DETAIL	AREA	Varchar2	25	-	-	-	✓	-	-
	ZONE	Varchar2	25	-	-	-	✓	-	-
									1 - 2

Fig-6.11: Detail table and Description.

12. **Create table Event:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Event (E_ID number (5) primary key, E_Name varchar (30), Location varchar (30))
Describe Event
```

Object Type **TABLE** Object **EVENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EVENT	E_ID	Number	-	5	0	1	-	-	-
	E_NAME	Varchar2	30	-	-	-	✓	-	-
	LOCATION	Varchar2	30	-	-	-	✓	-	-
									1 - 3

Fig-6.12: Event table and Description.

13. **Create table Moderator:**

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
create table Moderator (O_ID number (5) primary key, E_ID number (5),
          constraint efk foreign key (E_ID) references  Event (E_ID))
Describe Moderator
```

Object Type **TABLE** Object **MODERATOR**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MODERATOR	O_ID	Number	-	5	0	1	-	-	-
	E_ID	Number	-	5	0	-	✓	-	-
									1 - 2

Fig-6.13: Moderator table and Description.

14. **Create table Operates:**

User: AIUBCRICKET

Home > SQL > SQL Commands

☑ Autocommit Display 10

```
create table Operates (T_No number (4) primary key, Test varchar (20), ODI varchar (20),
              T20 varchar (20),Stadium varchar (20), Gate varchar (20),E_ID number (5),
              constraint oefk foreign key (E_ID) references  Event(E_ID))

Describe Operates
```

Object Type **TABLE** Object **OPERATES**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
OPERATES	T_NO	Number	-	4	0	1	-	-	-
	TEST	Varchar2	20	-	-	-	✓	-	-
	ODI	Varchar2	20	-	-	-	✓	-	-
	T20	Varchar2	20	-	-	-	✓	-	-
	STADIUM	Varchar2	20	-	-	-	✓	-	-
	GATE	Varchar2	20	-	-	-	✓	-	-
	E_ID	Number	-	5	0	-	✓	-	-
									1 - 7

Fig-6.14: Operates table and Description.

DATA INSERTION

1. **Value Insert with Player Table:**

P_ID	P_NAME	P_POSITION	P_DOB	P_AGE	SKILL	ROLE	CITY
49373	Bijoy Ahamed	Batsman	2000-02-15	23-Year	Right-Handed-Batsman	Captain	Brahmanbaria
49375	Anamul Haque	Wicket-Keeper	1990-09-18	34-Year	Left-Handed-Batsman	Wicket-Keeper	Dhaka
49377	Sakib Al Hasan	All-Rounder	1985-05-07	38-Year	All-Rounder	All-Rounder	Magura
49379	Taskin Ahamed	Bowler	1995-12-27	28-Year	First-Bowler	Bowler	Dhaka

Fig-7.1: Table of Player.

2. **Value Insert with Selector Table:**

S_ID	S_NAME
189	Md.Sajid Bin Faisal
269	Mohaimen-Bin-Noor
809	Mir MD.Kawsur

Fig-7.2: Table of Selector.

3. **Value Insert with Confirm Table:**

P_ID	S_ID
49373	189
49375	269
49377	809
49379	269

Fig-7.3: Table of Confirm.

4. **Value Insert with Address Table:**

COUNTRY	CITY
Bangladesh	Brahmanbaria
Bangladesh	Dhaka
Bangladesh	Magura
Bangladesh	Dhaka

Fig-7.4: Table of Address.

5. **Value Insert with Official Table:**

O_ID	OP_NAME	JOINING	JOB	ZONE
373	Abid Ahamed	2019-06-15	Director	Gulshan-2
375	Mahid Haque	2020-02-11	Financial	Banani-7
377	MD.Hasan	2022-01-09	Equipment Manager	Uttara-5
379	Azim Ahamed	2019-12-22	Stadium Manager	Dhanmondi-1

Fig-7.5: Table of Official.

6. **Value Insert with Selects Table:**

P_ID	O_ID
49373	373
49375	375
49377	377
49379	379

Fig-7.6: Table of Selects.

7. **Value Insert with Detail_Address Table:**

AREA	ZONE	COUNTRY	CITY
Gulshan	Gulshan-2	Bangladesh	Brahmanbaria
Banani	Banani-7	Bangladesh	Dhaka
Uttara	Uttara-5	Bangladesh	Magura
Dhanmondi	Dhanmondi-1	Bangladesh	Dhaka

Fig-7.7: Table of Detail_Address.

8. **Value Insert with Coache Table:**

C_ID	C_NAME	C_ROLE	C_E_MAIL
23	Ashraful Islam	Batting Coache	ashraful@gmail.com
24	Mashrafi Bin Murtoza	Fielding	mash202@gmail.com
25	Abdur Razzak	Bowling	razzak@gmail.com

Fig-7.8: Table of Coache.

9. **Value Insert with Guid Table:**

P_ID	C_ID
49373	23
49375	24
49377	25
49379	-

Fig-7.9: Table of Guid.

10. **Value Insert with Appoint Table:**

O_ID	C_ID
373	23
375	24
377	25
379	24

Fig-7.10: Table of Appoint.

11. **Value Insert with Detail Table:**

AREA	ZONE
Gulshan	Gulshan-2
Banani	Banani-7
Uttara	Uttara-5
Dhanmondi	Dhanmondi-1

Fig-7.11: Table of Detail.

12. Value Insert with Event Table:

E_ID	E_NAME	LOCATION
3396	ACL	Kuratuli-Dhaka
2572	PUCL	Mirpur Stadium
5396	BSMRCL	Sylhet Stadium

Fig-7.12: Table of Event.

13. Value Insert with Moderator Table:

O_ID	E_ID
373	3396
375	2572
377	5396
379	3396

Fig-7.13: Table of Moderator.

14. Value Insert with Operates Table:

T_NO	TEST	ODI	T20	STADIUM	GATE	E_ID
1001	1	2	3	MC Stadium	Gate A	2572
1002	2	-	3	AC Stadium	VIP Gate	3396
1003	-	2	3	SC Stadium	Gate 3	5396
1004	3	3	2	SHC Stadium	West Gate	3396

Fig-7.14: Table of Operates.

QUERY TEST

- **Simple query:**
  - Show all the associated information of the Player named "Bijoy Ahamed" from Player table?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT *FROM Player
WHERE  P_NAME ='Bijoy Ahamed';
```

P_ID	P_NAME	P_POSITION	P_DOB	P_AGE	SKILL	ROLE	CITY
49373	Bijoy Ahamed	Batsman	2000-02-15	23-Year	Right-Handed-Batsman	Captain	Brahmanbaria

Fig-8.1: Result of the Simple Query.

- **Query with a Single row function:**
  - Write a Single row function from official table and Show the Official ID, joining, job and zone of the official person where the job of the official person is 'DIRECTOR'?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT O_ID, joining, job, zone FROM Official
WHERE UPPER(job)='DIRECTOR';
```

O_ID	JOINING	JOB	ZONE
373	2019-06-15	Director	Gulshan-2

Fig-8.2: Result of the Single row function Query.

- **Query with a Multiple row function/ aggregate function:**
  - Show the total Coaches ID from Coache table?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT COUNT(C_ID) FROM Coache
```

COUNT(C_ID)
3

Fig-8.3: Result of the Aggregate function Query.

- **Subquery:**

a. Single row subquery:

➢ Who gets the p_id less than 'taskin ahamed' from player table?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT P_ID, P_Name FROM Player
WHERE P_ID < (Select P_ID from Player Where LOWER(P_Name) = 'taskin ahamed')
```

P_ID	P_NAME
49373	Bijoy Ahamed
49375	Anamul Haque
49377	Sakib Al Hasan

Fig-8.4a: Result of the Single row subquery.

b. Multiple row subquery:

➢ Show the name, joining date, job and zone of the official person from official table. Who gets the official id less than any of the 'Equipment Manager' and his job is not 'Equipment Manager?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT OP_Name, Joining, Job, Zone FROM Official
WHERE O_ID < ANY (Select O_ID from Official Where job = 'Equipment Manager') AND JOB <> 'Equipment Manager';
```

OP_NAME	JOINING	JOB	ZONE
Abid Ahamed	2019-06-15	Director	Gulshan-2
Mahid Haque	2020-02-11	Financial	Banani-7

Fig-8.4b: Result of the Multiple row subquery.

- **Joining:**

a. Equijoin:

- Join a table from Official and Detail table taking the Official Person name, job, area, and zone?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT O.OP_Name, O.Job, D.Area, O.Zone, D.Zone
FROM Official O, Detail D
WHERE O.Zone = D.Zone
```

OP_NAME	JOB	AREA	ZONE	ZONE
Abid Ahamed	Director	Gulshan	Gulshan-2	Gulshan-2
Mahid Haque	Financial	Banani	Banani-7	Banani-7
MD.Hasan	Equipment Manager	Uttara	Uttara-5	Uttara-5
Azim Ahamed	Stadium Manager	Dhanmondi	Dhanmondi-1	Dhanmondi-1

Fig-8.5a: Result of the Equijoin Query.

b. Outer Join:

- Join a table from Event and Operates table taking the E_Name, E_ID and Stadium. If a E_Name has a null display it has "NO NAME"?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
SELECT NVL(E.E_Name, 'NO NAME') AS E_Name, O.E_ID, O.Stadium FROM Event E, Operates O
WHERE O.E_ID  = E.E_ID(+)
```

E_NAME	E_ID	STADIUM
PUCL	2572	MC Stadium
ACL	3396	AC Stadium
BSMRCL	5396	SC Stadium
ACL	3396	SHC Stadium

Fig-8.5b: Result of the Outer join Query.

- **Create View:**

  a. Simple View:

  - Create a view named as Oview377 where the official id 377 will be shown over the columns O_ID, OP_Name, Joining and Job from Official table?

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
Create view Oview377 as Select O_ID, OP_Name, Joining, Job From Official
Where O_ID = 377;

Describe Oview377

Select * from Oview377
```

Object Type **VIEW** Object **OVIEW377**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
OVIEW377	O_ID	Number	-	5	0	-	-	-	-
	OP_NAME	Varchar2	25	-	-	-	✓	-	-
	JOINING	Varchar2	25	-	-	-	✓	-	-
	JOB	Varchar2	25	-	-	-	✓	-	-
									1 - 4

O_ID	OP_NAME	JOINING	JOB
377	MD.Hasan	2022-01-09	Equipment Manager

Fig-8.6a: Result of the Simple View Query.

b. Complex View:

- Create a view name "CE" to show the joining between the tables Coache and Appoint.

User: AIUBCRICKET

Home > SQL > SQL Commands

Autocommit Display 10

```
Create View CE(Coache_name, Coache_role, Official_ID) as Select C.C_Name, C.C_Role, A.O_ID
FROM Coache C, Appoint A
WHERE C.C_ID(+) = A.C_ID;

DESCRIBE CE

SELECT *FROM CE
```

Object Type **VIEW** Object **CE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CE	COACHE_NAME	Varchar2	20	-	-	-	✓	-	-
	COACHE_ROLE	Varchar2	20	-	-	-	✓	-	-
	OFFICIAL_ID	Number	-	5	0	-	-	-	-

1 - 3

COACHE_NAME	COACHE_ROLE	OFFICIAL_ID
Ashraful Islam	Batting Coache	373
Mashrafi Bin Murtoza	Fielding	375
Abdur Razzak	Bowling	377
Mashrafi Bin Murtoza	Fielding	379

Fig-8.6b: Result of the Complex View Query.

DB CONNECTION

1. Mysql java connector (jar file)
   You can go to **"myscql java connector maven" any version you like.**
2. You need to search (xampp apache mariadb perl php) install xampp server normally.
3. Select the xampp icon control and then start both apache and mysql. Next go to mysql admin panel.
4. Now create a Database, inside of it create table and insert the values in the table. The table in mysql is:

E_ID	E_NAME	LOCATION
3396	ACL	Kuratoli-Dhaka
2572	PUCL	Mirpur Stadium
5396	BSMRCL	Sylhet Stadium

Fig-9.1: Result of the mysql table.

Now we need an IDE.
Before that, we need to understand the steps of DB connection.

- Register Driver

```
Class.forName( className: "com.mysql.jdbc.Driver");
```

- Connect the DB

```
Connection con = DriverManager.getConnection(
        url: "jdbc:mysql://localhost:3306/player",  user: "root",  password: "");
```

- Create statement

```
Statement stmt = con.createStatement();
```

- Execute the query () in the resultstmnt

```
ResultSet rs = stmt.executeQuery( sql: "select * from event");
```

The while loop iterates through the rows of the result set (rs). The next() method advances the cursor to the next row in the result set and returns true if there is a next row, and false if there are no more rows.

```
while (rs.next())
    System.out.println(rs.getInt( columnIndex: 1) + "  " + rs.getString( columnIndex: 2)
            + "  " + rs.getString( columnIndex: 3));
```

- Connection close ()

```
con.close();
```

5. Now write the DB connection code and add the jar file location in the libraries. The output will be

```
3396  ACL  Kuratoli-Dhaka
2572  PUCL  Mirpur Stadium
5396  BSMRCL  Sylhet Stadium
```

Fig-9.1: Result of the output table.

CONCLUSION

Our project on AIUB Cricket Team Management is built so that it can be used as a database system of AIUB cricket team management system. We tried to include all types of entities and store their information in the possible easiest way. Through this project, we have put our theoretical knowledge into a practical project that was learnt from this Introduction to Database course. The knowledge we gathered throughout this course and this project hopefully will help to build any projects related to databases we will have in the coming future. We are also very excited to work on this project and willing to add more features as well.